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前言

概述

本文档描述RK3568 SATA性能测试和指标数据，测试基于RK3568 EVB1 开发板。

各芯片 **feature** 支持状态

芯片名称	内核版本
RK3568/RK3566	4.19

读者对象

本文档主要适用于以下工程师：

技术支持工程师

软件开发工程师

项目管理人员

测试工程师

修订记录

版本号	作者	修改日期	修改说明
V1.00	赵仪峰	2020-12-21	初始版本

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1. SATA硬盘测试方法

固态硬盘每次测试写数据前建议先格式化一下磁盘或者删除测试文件，让固态硬盘恢复写性能，命令如下：

```
mkfs.ext4 /dev/block/sda  
rm /mnt/sda/test
```

- **dd 速率统计：** 使用带有速率计算功能的 dd 命令进行统计；
- 使用格式化命令进行格式化， 如下：

```
mkdosfs -F 32 /dev/block/sda  
mkfs.ext4 /dev/block/sda
```

- 挂载硬盘到/mnt/目录， 例如：

```
mkdir /mnt/sda  
mount -t vfat /dev/block/sda /mnt/sda  
mount -t ext4 /dev/block/sda /mnt/sda
```

- 写速率：

```
dd if=/dev/zero of=/mnt/sda/test bs=1024K count=2048
```

- 读速率：

```
echo 3 > /proc/sys/vm/drop_caches  
dd if=/mnt/sda/test of=/dev/null bs=1024K count=2048
```

- **FIO RAW写性能测试：**

```
fio -filename=/dev/block/sda -direct=1 -iodepth 1 -thread -rw=write -  
ioengine=psync -bs=1024k -size=10G -numjobs=10 -runtime=180 -group_reporting  
-name=seq_100write_1024k
```

- **FIO 写性能测试：**

```
fio -filename=/mnt/sda/test -direct=1 -iodepth 1 -thread -rw=write -  
ioengine=psync -bs=1024k -size=10G -numjobs=10 -runtime=180 -group_reporting  
-name=seq_100write_1024k
```

- **FIO RAW读性能测试：**

```
fio -filename=/dev/block/sda -direct=1 -iodepth 1 -thread -rw=read -  
ioengine=psync -bs=1024k -size=10G -numjobs=10 -runtime=180 -group_reporting  
-name=seq_100read_1024k
```

- FIO 读性能测试:

```
fio -filename=/mnt/sda/test -direct=1 -iodepth 1 -thread -rw=read -
ioengine=psync -bs=1024k -size=10G -numjobs=10 -runtime=180 -group_reporting
-name=seq_100read_1024k
```

- FIO RAW读写性能测试:

```
fio -filename=/dev/block/sda -direct=1 -iodepth 1 -thread -rw=randrw -
rwmixread=70 -ioengine=psync -bs=128k -size=30G -numjobs=10 -runtime=180 -
group_reporting -name=randrw_70read_128k
```

- FIO 读写性能测试:

```
fio -filename=/mnt/sda/test -direct=1 -iodepth 1 -thread -rw=randrw -
rwmixread=70 -ioengine=psync -bs=128k -size=30G -numjobs=10 -runtime=180 -
group_reporting -name=randrw_70read_128k
```

- 1.5G、3.0G 和 6.0G 速率配置:

```
6G: io -w -4 0xfdc50000 0x33302220
3G: io -w -4 0xfdc50000 0x33301110
1.5G: io -w -4 0xfdc50000 0x33300000
```

2. SATA机械硬盘测试性能

2.1 机械硬盘测试环境

型号	品牌	容量	模式	备注
WD10EZEX	WD	1TB	SATA1.0&2.0&3.0	64MB 缓存, 蓝 盘

2.2 机械硬盘测试数据

注: 由于FIO测试命令采用10个线程读写同一个文件, 部分读取会直接从硬盘内部缓存返回, 读速度比单线读取要快很多。写数据时多线程会比单线程慢一点。

2.2.1 1.5G测试数据

2.2.1.1 机械硬盘DD测试

测试项	FAT32	EXT4
DD写	79MB/S	130MB/S
DD读	135MB/S	117MB/S

2.2.1.2 机械硬盘FIO测试

测试项	RAW	EXT4
FIO写	118MB/S	91.2MB/S
FIO读	143MB/S	138MB/S
FIO读写	R:19.2MB/S W:8523KB/S	R:11.1MB/S W:4715KB/S

2.2.2 3G测试数据

2.2.2.1 机械硬盘DD测试

测试项	FAT32	EXT4
DD写	98MB/S	174MB/S
DD读	186MB/S	176MB/S

2.2.2.2 机械硬盘FIO测试

测试项	RAW	EXT4
FIO写	193MB/S	163MB/S
FIO读	276MB/S	262MB/S
FIO读写	R:20.3MB/S W:8601KB/S	R:12.7MB/S W:5447KB/S

2.2.3 6G测试数据

2.2.3.1 机械硬盘DD测试

测试项	FAT32	EXT4
DD写	102MB/S	176MB/S
DD读	187MB/S	177MB/S

2.2.3.2 机械硬盘FIO测试

测试项	RAW	EXT4
FIO写	238MB/S	188MB/S
FIO读	451MB/S	463MB/S
FIO读写	R:19.3MB/S W:8567KB/S	R:11.7MB/S W:4967KB/S

3. 固态硬盘测试性能

3.1 固态硬盘测试环境

型号	品牌	容量	模式	备注
SC001	长江存储	256GB	SATA1.0&2.0&3.0	FW: YM013200

3.2 固态硬盘测试数据

3.2.1 1.5G测试数据

3.2.1.1 固态硬盘DD测试

测试项	FAT32	EXT4
DD写	83MB/S	135MB/S
DD读	135MB/S	129MB/S

3.2.1.2 固态硬盘FIO测试

测试项	RAW	EXT4
FIO写	132MB/S	110MB/S
FIO读	142MB/S	142MB/S
FIO读写	R:95.1MB/S W:41.2MB/S	R:72.1MB/S W:30.8MB/S

3.2.2 3G测试数据

3.2.2.1 固态硬盘DD测试

测试项	FAT32	EXT4
DD写	122MB/S	251MB/S
DD读	267MB/S	257MB/S

3.2.2.2 固态硬盘FIO测试

测试项	RAW	EXT4
FIO写	257MB/S	197MB/S
FIO读	284MB/S	284MB/S
FIO读写	R:189MB/S W:81.2MB/S	R: 121MB/S W:57MB/S

3.2.3 6G测试数据

3.2.3.1 固态硬盘DD测试

测试项	FAT32	EXT4
DD写	162MB/S	287MB/S
DD读	403MB/S	443MB/S

3.2.3.2 固态硬盘FIO测试

测试项	RAW	EXT4
FIO写	378MB/s	313MB/S
FIO读	567MB/s	547MB/S
FIO读写	R: 339MB/S W:145MB/S	R:175MB/S W:74.9MB/S

4. SATA 硬盘测试结论

4.1 机械硬盘性能

测试了型号为 WD10EZEX 的一款 1TB SATA 机械硬盘在 fat32和ext4 文件系统下DD读写性能，以及ext4 文件系统和无文件系统下FIO的读写性能。

4.1.1 DD读写性能

FAT32文件系统下，读速率在1.5Gbps模式可以达到135MB/S， 3.0Gbps 和 6.0Gbps 模式下可达到186MB/s。

EXT4文件系统下，写速率在1.5Gbps模式可以达到130MB/S， 3.0Gbps 和 6.0Gbps 模式下可达到174MB/s。

1.5Gbps模式时速率接近接口速率，3.0Gbps 和 6.0Gbps 模式时速率接近硬盘最高速率。

4.1.2 FIO读写性能

多线程读时，在6Gbps模式下读可以达到451MB/S,超过了机械硬盘实际速率，通过协议分析仪抓取数据，确定是多线程读取同一个文件，大部分读会命中硬盘的缓存。

多线程写时，在6Gbps模式下读可以达到180MB/S, 接近硬盘写性能。

多线程读写并发时，读写性能下降非常多，符合机械硬盘特性。详情可以参考测试原始数据。

4.1.3 1.5G、3.0G 和 6.0G 模式对比

6.0Gbps、3.0Gbps、1.5Gbps 的性能依次降低，但是写速率差别不是很显著，读速率 3.0Gbps 和 6.0Gbps 模式下显著高于 1.5Gbps。

4.1.4 fat32和ext4 文件系统对比

从整体来看，写性能在 ext4文件系统下显著高于 fat32，读性能相差不大。

4.2 固态硬盘性能

测试了型号为 SC001的一款 256GB SATA 固态硬盘（64层3D TLC）在 fat32和ext4 文件系统下DD读写性能，以及ext4文件系统和无文件系统下FIO的读写性能。

4.2.1 DD读写性能

FAT32文件系统下，读速率在1.5Gbps模式可以达到135MB/S， 3.0Gbps模式下可达到 267MB/s，6.0Gbps模式下可以达到407MB/S 。

EXT4文件系统下，写速率在1.5Gbps模式可以达到135MB/S， 3.0Gbps模式下可达到 251MB/s， 6.0Gbps 模式下可以达到287MB/S 。

1.5Gbps和3.0Gbps模式时速率接近接口速率，6.0Gbps 模式时速率接近硬盘单线程最高速率。

4.2.2 FIO读写性能

RAW模式下读写性能比EXT4文件系统要好一点，多线程读时，在6Gbps模式下读可以达到567MB/S,接近接口速率。

多线程写时，在6Gbps模式下读可以达到378MB/S, 接近硬盘写性能。在3Gbps和1.5Gbps模式下，速率接近接口速率。

多线程读写时，在EXT4文件系统下，读写性能下降比较多，原因是文件系统小数据读写增加。无文件系统下读写速度加起来接近接口速率。

4.2.3 1.5G、3.0G 和 6.0G 模式对比

从整体来看，1.5Gbps和3.0Gbps模式下读写性能都接近接口速率，6.0Gbps 的读写性能显著优于1.5Gbps和3.0Gbps模式性能。

4.2.4 fat32和ext4 文件系统对比

从整体来看，写性能在 ext4文件系统下显著高于 fat32，读性能相差不大。

5. 测试原始数据

5.1 6G 固态FIO RAW写

```
seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1874: Tue Nov 24 18:01:16
2020
write: IOPS=360, BW=361MiB/s (378MB/s) (63.4GiB/180053msec)
  clat (msec): min=6, max=499, avg=27.40, stdev=23.79
  lat (msec): min=6, max=499, avg=27.69, stdev=23.79
  clat percentiles (msec):
    | 1.00th=[ 20], 5.00th=[ 20], 10.00th=[ 20], 20.00th=[ 20],
    | 30.00th=[ 20], 40.00th=[ 20], 50.00th=[ 20], 60.00th=[ 20],
    | 70.00th=[ 20], 80.00th=[ 20], 90.00th=[ 58], 95.00th=[ 59],
    | 99.00th=[ 61], 99.50th=[ 89], 99.90th=[ 306], 99.95th=[ 338],
    | 99.99th=[ 498]
  bw ( KiB/s): min= 4104, max=54113, per=0.01%, avg=37206.85, stdev=15908.21
  lat (msec) : 10=0.01%, 20=80.55%, 50=2.84%, 100=16.19%, 250=0.03%
  lat (msec) : 500=0.38%
cpu           : usr=1.09%, sys=0.98%, ctx=68607, majf=0, minf=0
IO depths     : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=0,64943,0, short=0,0,0, dropped=0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=361MiB/s (378MB/s), 361MiB/s-361MiB/s (378MB/s-378MB/s), io=63.4GiB
(68.1GB), run=180053-180053msec

Disk stats (read/write):
  sda: ios=0/64908, merge=0/0, ticks=0/1759057, in_queue=1757523, util=100.00%
```

5.2 6G 固态FIO RAW读

```
seq_100read_1024k: (g=0): rw=read, bs=(R) 1024KiB-1024KiB, (W) 1024KiB-1024KiB,
(T) 1024KiB-1024KiB, ioengine=psync, iodepth=1
...
fio-2.20
Starting 10 threads
Jobs: 10 (f=10): [R(10)][100.0%][r=541MiB/s,w=0KiB/s][r=541,w=0 IOPS][eta
00m:00s]
seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=1795: Tue Nov 24 17:37:13
2020
    read: IOPS=540, BW=541MiB/s (567MB/s) (31.7GiB/60018msec)
        clat (usec): min=16345, max=27735, avg=18463.75, stdev=102.66
        lat (usec): min=16348, max=27737, avg=18466.33, stdev=102.57
        clat percentiles (usec):
            | 1.00th=[18304], 5.00th=[18304], 10.00th=[18304], 20.00th=[18304],
            | 30.00th=[18560], 40.00th=[18560], 50.00th=[18560], 60.00th=[18560],
            | 70.00th=[18560], 80.00th=[18560], 90.00th=[18560], 95.00th=[18560],
            | 99.00th=[18560], 99.50th=[18560], 99.90th=[18816], 99.95th=[18816],
            | 99.99th=[23936]
        bw ( KiB/s): min=53354, max=58040, per=0.01%, avg=55715.52, stdev=505.47
        lat (msec) : 20=99.97%, 50=0.03%
    cpu          : usr=0.19%, sys=1.69%, ctx=68267, majf=0, minf=2560
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
        submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
        complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
        issued rwts: total=32450,0,0, short=0,0,0, dropped=0,0,0
        latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
    READ: bw=541MiB/s (567MB/s), 541MiB/s-541MiB/s (567MB/s-567MB/s), io=31.7GiB
(34.0GB), run=60018-60018msec

Disk stats (read/write):
    sda: ios=35595/0, merge=0/0, ticks=642173/0, in_queue=635899, util=100.00%
```

5.3 6G 固态FIO RAW读写

```
randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=2159: Tue Nov 24 21:58:19
2020
    read: IOPS=2589, BW=324MiB/s (339MB/s) (56.9GiB/180003msec)
        clat (usec): min=602, max=188853, avg=1267.86, stdev=937.61
        lat (usec): min=605, max=188854, avg=1269.76, stdev=937.65
        clat percentiles (usec):
            | 1.00th=[ 684], 5.00th=[ 732], 10.00th=[ 900], 20.00th=[ 932],
            | 30.00th=[ 980], 40.00th=[ 1096], 50.00th=[ 1176], 60.00th=[ 1256],
            | 70.00th=[ 1368], 80.00th=[ 1496], 90.00th=[ 1704], 95.00th=[ 1896],
            | 99.00th=[ 4048], 99.50th=[ 4640], 99.90th=[ 5728], 99.95th=[ 6816],
            | 99.99th=[21376]
        bw ( KiB/s): min=16128, max=49250, per=0.01%, avg=33222.29, stdev=3878.80
    write: IOPS=1109, BW=139MiB/s (145MB/s) (24.4GiB/180003msec)
        clat (usec): min=496, max=196193, avg=5944.85, stdev=2379.55
        lat (usec): min=520, max=196229, avg=5974.87, stdev=2379.22
        clat percentiles (usec):
            | 1.00th=[ 2800], 5.00th=[ 3408], 10.00th=[ 3792], 20.00th=[ 4320],
            | 30.00th=[ 4768], 40.00th=[ 5216], 50.00th=[ 5664], 60.00th=[ 6112],
            | 70.00th=[ 6624], 80.00th=[ 7328], 90.00th=[ 8384], 95.00th=[ 9280],
```

```

    | 99.00th=[11456], 99.50th=[12608], 99.90th=[24960], 99.95th=[26496],
    | 99.99th=[29568]
bw (   KiB/s): min= 7951, max=17255, per=0.01%, avg=14234.57, stdev=963.78
  lat (usec) : 500=0.01%, 750=4.07%, 1000=18.99%
  lat (msec) : 2=44.47%, 4=5.78%, 10=25.78%, 20=0.82%, 50=0.08%
  lat (msec) : 250=0.01%
cpu           : usr=1.28%, sys=3.73%, ctx=1119108, majf=0, minf=0
IO depths     : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
  submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  issued rwt: total=466133,199726,0, short=0,0,0, dropped=0,0,0
  latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=324MiB/s (339MB/s), 324MiB/s-324MiB/s (339MB/s-339MB/s), io=56.9GiB
(61.1GB), run=180003-180003msec
  WRITE: bw=139MiB/s (145MB/s), 139MiB/s-139MiB/s (145MB/s-145MB/s), io=24.4GiB
(26.2GB), run=180003-180003msec

Disk stats (read/write):
  sda: ios=465784/199535, merge=0/0, ticks=550694/1168569, in_queue=1691665,
util=100.00%

```

5.4 6G 固态FIO EXT4写

```

seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=2258: Tue Nov 24 22:40:13
2020
  write: IOPS=298, BW=298MiB/s (313MB/s) (52.4GiB/180053msec)
    clat (msec): min=2, max=663, avg=33.19, stdev=17.39
    lat (msec): min=3, max=663, avg=33.49, stdev=17.39
    clat percentiles (msec):
      | 1.00th=[ 28], 5.00th=[ 30], 10.00th=[ 31], 20.00th=[ 31],
      | 30.00th=[ 31], 40.00th=[ 31], 50.00th=[ 31], 60.00th=[ 31],
      | 70.00th=[ 31], 80.00th=[ 33], 90.00th=[ 34], 95.00th=[ 46],
      | 99.00th=[ 71], 99.50th=[ 196], 99.90th=[ 258], 99.95th=[ 281],
      | 99.99th=[ 660]
    bw (   KiB/s): min= 2052, max=36937, per=0.01%, avg=30764.35, stdev=4778.12
    lat (msec) : 4=0.05%, 10=0.01%, 20=0.01%, 50=95.77%, 100=3.52%
    lat (msec) : 250=0.53%, 500=0.09%, 750=0.02%
    cpu           : usr=0.75%, sys=4.72%, ctx=109396, majf=0, minf=3
    IO depths     : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
      submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      issued rwt: total=0,53699,0, short=0,0,0, dropped=0,0,0
      latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=298MiB/s (313MB/s), 298MiB/s-298MiB/s (313MB/s-313MB/s), io=52.4GiB
(56.3GB), run=180053-180053msec

Disk stats (read/write):
  sda: ios=1/63043, merge=0/268, ticks=1/189794, in_queue=192985, util=84.22%

```

5.5 6G 固态FIO EXT4读

```
seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=2284: Tue Nov 24 22:47:02
2020
  read: IOPS=521, BW=521MiB/s (547MB/s) (91.7GiB/180019msec)
    clat (msec): min=6, max=767, avg=19.15, stdev= 8.45
    lat (msec): min=6, max=767, avg=19.15, stdev= 8.45
    clat percentiles (msec):
      | 1.00th=[ 19], 5.00th=[ 19], 10.00th=[ 19], 20.00th=[ 19],
      | 30.00th=[ 19], 40.00th=[ 19], 50.00th=[ 19], 60.00th=[ 19],
      | 70.00th=[ 19], 80.00th=[ 19], 90.00th=[ 19], 95.00th=[ 20],
      | 99.00th=[ 37], 99.50th=[ 48], 99.90th=[ 65], 99.95th=[ 74],
      | 99.99th=[ 742]
    bw ( KiB/s): min=14336, max=58040, per=0.01%, avg=53712.45, stdev=4266.89
    lat (msec) : 10=0.01%, 20=95.02%, 50=4.53%, 100=0.43%, 250=0.01%
    lat (msec) : 750=0.01%, 1000=0.01%
  cpu          : usr=0.18%, sys=4.25%, ctx=95144, majf=0, minf=2560
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwt: total=93858,0,0, short=0,0,0, dropped=0,0,0
    latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=521MiB/s (547MB/s), 521MiB/s-521MiB/s (547MB/s-547MB/s), io=91.7GiB
(98.4GB), run=180019-180019msec

Disk stats (read/write):
  sda: ios=103230/486, merge=5/11, ticks=1915566/84958, in_queue=1976259,
util=100.00%
```

5.6 6G 固态FIO EXT4读写

```
randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=2314: Tue Nov 24 22:56:26
2020
  read: IOPS=1332, BW=167MiB/s (175MB/s) (29.3GiB/180006msec)
    clat (msec): min=1, max=41, avg= 5.15, stdev= 1.86
    lat (msec): min=1, max=41, avg= 5.15, stdev= 1.86
    clat percentiles (usec):
      | 1.00th=[ 2008], 5.00th=[ 2800], 10.00th=[ 3408], 20.00th=[ 3984],
      | 30.00th=[ 4448], 40.00th=[ 4768], 50.00th=[ 5024], 60.00th=[ 5344],
      | 70.00th=[ 5664], 80.00th=[ 6048], 90.00th=[ 6624], 95.00th=[ 7328],
      | 99.00th=[10176], 99.50th=[11840], 99.90th=[27008], 99.95th=[31872],
      | 99.99th=[37120]
    bw ( KiB/s): min=10773, max=22573, per=0.01%, avg=17081.38, stdev=1432.00
  write: IOPS=571, BW=71.4MiB/s (74.9MB/s) (12.5GiB/180006msec)
    clat (usec): min=485, max=40867, avg=5355.29, stdev=1888.80
    lat (usec): min=515, max=40904, avg=5395.19, stdev=1887.50
    clat percentiles (usec):
      | 1.00th=[ 2672], 5.00th=[ 3344], 10.00th=[ 3824], 20.00th=[ 4256],
      | 30.00th=[ 4640], 40.00th=[ 4960], 50.00th=[ 5216], 60.00th=[ 5472],
      | 70.00th=[ 5792], 80.00th=[ 6176], 90.00th=[ 6688], 95.00th=[ 7584],
      | 99.00th=[10304], 99.50th=[12352], 99.90th=[28288], 99.95th=[33024],
```

```

| 99.99th=[38144]
bw ( KiB/s): min= 2816, max=11286, per=0.01%, avg=7322.25, stdev=1135.78
lat (usec) : 500=0.01%, 750=0.18%, 1000=0.04%
lat (msec) : 2=0.70%, 4=17.46%, 10=80.40%, 20=0.95%, 50=0.27%
cpu          : usr=0.98%, sys=6.38%, ctx=682432, majf=0, minf=0
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=239797,102799,0, short=0,0,0, dropped=0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=167MiB/s (175MB/s), 167MiB/s-167MiB/s (175MB/s-175MB/s), io=29.3GiB
(31.4GB), run=180006-180006msec
  WRITE: bw=71.4MiB/s (74.9MB/s), 71.4MiB/s-71.4MiB/s (74.9MB/s-74.9MB/s),
io=12.5GiB (13.5GB), run=180006-180006msec

Disk stats (read/write):
  sda: ios=239570/106858, merge=0/78, ticks=287546/59409, in_queue=331198,
util=75.19%

```

5.7 3G 固态FIO RAW写

```

seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1724: Tue Nov 24 15:28:01
2020
write: IOPS=245, BW=246MiB/s (257MB/s) (43.2GiB/180036msec)
clat (msec): min=8, max=409, avg=40.40, stdev=18.29
lat (msec): min=8, max=409, avg=40.69, stdev=18.29
clat percentiles (msec):
| 1.00th=[ 36], 5.00th=[ 36], 10.00th=[ 39], 20.00th=[ 39],
| 30.00th=[ 39], 40.00th=[ 39], 50.00th=[ 39], 60.00th=[ 39],
| 70.00th=[ 39], 80.00th=[ 39], 90.00th=[ 40], 95.00th=[ 40],
| 99.00th=[ 60], 99.50th=[ 128], 99.90th=[ 322], 99.95th=[ 322],
| 99.99th=[ 338]
bw ( KiB/s): min= 6193, max=29138, per=0.01%, avg=25342.06, stdev=3108.25
lat (msec) : 10=0.01%, 20=0.01%, 50=96.71%, 100=2.64%, 250=0.26%
lat (msec) : 500=0.38%
cpu          : usr=0.73%, sys=0.96%, ctx=46661, majf=0, minf=3
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=0,44207,0, short=0,0,0, dropped=0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=246MiB/s (257MB/s), 246MiB/s-246MiB/s (257MB/s-257MB/s), io=43.2GiB
(46.4GB), run=180036-180036msec

Disk stats (read/write):
  sda: ios=0/52957, merge=0/0, ticks=0/2106071, in_queue=2105193, util=100.00%

```

5.8 3G 固态FIO RAW读

```

seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=1701: Tue Nov 24 15:25:18
2020
  read: IOPS=270, BW=271MiB/s (284MB/s) (47.6GiB/180034msec)
    clat (usec): min=20181, max=44303, avg=36903.07, stdev=139.02
    lat (usec): min=20183, max=44304, avg=36905.65, stdev=138.95
    clat percentiles (usec):
      | 1.00th=[36608], 5.00th=[36608], 10.00th=[36608], 20.00th=[37120],
      | 30.00th=[37120], 40.00th=[37120], 50.00th=[37120], 60.00th=[37120],
      | 70.00th=[37120], 80.00th=[37120], 90.00th=[37120], 95.00th=[37120],
      | 99.00th=[37120], 99.50th=[37120], 99.90th=[37120], 99.95th=[37120],
      | 99.99th=[38144]
    bw ( KiB/s): min=26624, max=29079, per=0.01%, avg=27866.93, stdev=1036.09
    lat (msec) : 50=100.00%
  cpu           : usr=0.10%, sys=1.02%, ctx=112243, majf=0, minf=2564
  IO depths     : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwt  : total=48741,0,0, short=0,0,0, dropped=0,0,0
    latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=271MiB/s (284MB/s), 271MiB/s-271MiB/s (284MB/s-284MB/s), io=47.6GiB
(51.1GB), run=180034-180034msec

Disk stats (read/write):
  sda: ios=63286/0, merge=0/0, ticks=2289137/0, in_queue=2279921, util=100.00%

```

5.9 3G 固态FIO RAW读写

```

randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=1721: Tue Nov 24 15:33:54
2020
  read: IOPS=1445, BW=181MiB/s (189MB/s) (31.8GiB/180005msec)
    clat (usec): min=1108, max=31177, avg=1525.27, stdev=421.16
    lat (usec): min=1108, max=31179, avg=1527.61, stdev=421.16
    clat percentiles (usec):
      | 1.00th=[ 1320], 5.00th=[ 1352], 10.00th=[ 1368], 20.00th=[ 1384],
      | 30.00th=[ 1384], 40.00th=[ 1400], 50.00th=[ 1416], 60.00th=[ 1416],
      | 70.00th=[ 1448], 80.00th=[ 1816], 90.00th=[ 1896], 95.00th=[ 1928],
      | 99.00th=[ 2384], 99.50th=[ 3824], 99.90th=[ 4896], 99.95th=[ 5344],
      | 99.99th=[16320]
    bw ( KiB/s): min= 8464, max=31294, per=0.01%, avg=18547.76, stdev=3244.61
  write: IOPS=619, BW=77.4MiB/s (81.2MB/s) (13.6GiB/180005msec)
    clat (msec): min=4, max=47, avg=12.46, stdev= 3.57
    lat (msec): min=4, max=47, avg=12.50, stdev= 3.57
    clat percentiles (usec):
      | 1.00th=[ 6240], 5.00th=[ 7648], 10.00th=[ 8256], 20.00th=[ 9536],
      | 30.00th=[10432], 40.00th=[11328], 50.00th=[11968], 60.00th=[12864],
      | 70.00th=[13760], 80.00th=[15168], 90.00th=[17024], 95.00th=[18816],
      | 99.00th=[22912], 99.50th=[24448], 99.90th=[29568], 99.95th=[32128],
      | 99.99th=[38144]
    bw ( KiB/s): min= 6168, max= 9490, per=0.01%, avg=7944.42, stdev=491.15
    lat (msec) : 2=68.78%, 4=0.91%, 10=8.30%, 20=21.08%, 50=0.92%
  cpu           : usr=0.87%, sys=2.60%, ctx=635887, majf=0, minf=0
  IO depths     : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%

```

```
complete : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=260174,111453,0, short=0,0,0, dropped=0,0,0
latency   : target=0, window=0, percentile=100.00%, depth=1
```

Run status group 0 (all jobs):

```
READ: bw=181MiB/s (189MB/s), 181MiB/s-181MiB/s (189MB/s-189MB/s), io=31.8GiB
(34.1GB), run=180005-180005msec
```

```
WRITE: bw=77.4MiB/s (81.2MB/s), 77.4MiB/s-77.4MiB/s (81.2MB/s-81.2MB/s),
io=13.6GiB (14.6GB), run=180005-180005msec
```

Disk stats (read/write):

```
sda: ios=259948/111345, merge=0/0, ticks=369841/1375732, in_queue=1725902,
util=100.00%
```

5.10 3G 固态FIO EXT4写

```
seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1764: Tue Nov 24 15:49:43
2020
```

```
write: IOPS=187, BW=188MiB/s (197MB/s) (33.0GiB/180050msec)
```

```
clat (msec): min=4, max=654, avg=52.85, stdev=18.20
```

```
lat (msec): min=4, max=655, avg=53.17, stdev=18.20
```

```
clat percentiles (msec):
```

```
| 1.00th=[ 48], 5.00th=[ 50], 10.00th=[ 50], 20.00th=[ 50],
| 30.00th=[ 50], 40.00th=[ 50], 50.00th=[ 50], 60.00th=[ 50],
| 70.00th=[ 50], 80.00th=[ 57], 90.00th=[ 57], 95.00th=[ 58],
| 99.00th=[ 72], 99.50th=[ 200], 99.90th=[ 326], 99.95th=[ 347],
| 99.99th=[ 652]
```

```
bw ( KiB/s): min= 2052, max=27002, per=0.01%, avg=19407.14, stdev=2083.78
```

```
lat (msec) : 10=0.07%, 20=0.01%, 50=70.37%, 100=28.93%, 250=0.41%
```

```
lat (msec) : 500=0.18%, 750=0.03%
```

```
cpu          : usr=1.04%, sys=2.36%, ctx=69291, majf=0, minf=3
```

```
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
```

```
submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
```

```
complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
```

```
issued rwt: total=0,33834,0, short=0,0,0, dropped=0,0,0
```

```
latency     : target=0, window=0, percentile=100.00%, depth=1
```

Run status group 0 (all jobs):

```
WRITE: bw=188MiB/s (197MB/s), 188MiB/s-188MiB/s (197MB/s-197MB/s), io=33.0GiB
(35.5GB), run=180050-180050msec
```

Disk stats (read/write):

```
sda: ios=1/40436, merge=0/244, ticks=1/204040, in_queue=192872, util=82.95%
```

5.11 3G 固态FIO EXT4读

```
seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=1797: Tue Nov 24 16:08:02
2020
```

```
read: IOPS=270, BW=271MiB/s (284MB/s) (47.6GiB/180035msec)
```

```
clat (usec): min=18046, max=57894, avg=36882.02, stdev=309.35
```

```
lat (usec): min=18049, max=57897, avg=36884.77, stdev=309.32
```

```
clat percentiles (usec):
```

```

| 1.00th=[36608], 5.00th=[36608], 10.00th=[36608], 20.00th=[37120],
| 30.00th=[37120], 40.00th=[37120], 50.00th=[37120], 60.00th=[37120],
| 70.00th=[37120], 80.00th=[37120], 90.00th=[37120], 95.00th=[37120],
| 99.00th=[37120], 99.50th=[37120], 99.90th=[37120], 99.95th=[37632],
| 99.99th=[54016]
bw ( KiB/s): min=26624, max=28787, per=0.01%, avg=27804.70, stdev=1022.87
lat (msec) : 20=0.01%, 50=99.98%, 100=0.01%
cpu          : usr=0.11%, sys=2.26%, ctx=49150, majf=0, minf=2560
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=48769,0,0, short=0,0,0, dropped=0,0,0
latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=271MiB/s (284MB/s), 271MiB/s-271MiB/s (284MB/s-284MB/s), io=47.6GiB
(51.1GB), run=180035-180035msec

Disk stats (read/write):
sda: ios=48745/8, merge=0/1, ticks=1771129/824, in_queue=1761614, util=100.00%
```

5.12 3G 固态FIO EXT4读写

```

randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=1866: Tue Nov 24 16:39:27
2020
read: IOPS=919, BW=115MiB/s (121MB/s) (20.2GiB/180009msec)
clat (usec): min=936, max=47371, avg=7464.51, stdev=2302.38
lat (usec): min=939, max=47374, avg=7467.06, stdev=2302.40
clat percentiles (usec):
| 1.00th=[ 3184], 5.00th=[ 4640], 10.00th=[ 5536], 20.00th=[ 6176],
| 30.00th=[ 6624], 40.00th=[ 6944], 50.00th=[ 7264], 60.00th=[ 7648],
| 70.00th=[ 7968], 80.00th=[ 8384], 90.00th=[ 9024], 95.00th=[10304],
| 99.00th=[15936], 99.50th=[17024], 99.90th=[30848], 99.95th=[35072],
| 99.99th=[42752]
bw ( KiB/s): min= 7182, max=15421, per=0.01%, avg=11792.56, stdev=1079.07
write: IOPS=394, BW=49.3MiB/s (51.7MB/s) (8879MiB/180009msec)
clat (usec): min=683, max=46903, avg=7791.02, stdev=2291.15
lat (usec): min=707, max=46939, avg=7828.04, stdev=2291.56
clat percentiles (usec):
| 1.00th=[ 5088], 5.00th=[ 5600], 10.00th=[ 6112], 20.00th=[ 6560],
| 30.00th=[ 6944], 40.00th=[ 7200], 50.00th=[ 7520], 60.00th=[ 7712],
| 70.00th=[ 8096], 80.00th=[ 8384], 90.00th=[ 9024], 95.00th=[11072],
| 99.00th=[16064], 99.50th=[17280], 99.90th=[34048], 99.95th=[38656],
| 99.99th=[44800]
bw ( KiB/s): min= 1285, max= 8481, per=0.01%, avg=5057.83, stdev=949.29
lat (usec) : 750=0.01%, 1000=0.05%
lat (msec) : 2=0.07%, 4=1.30%, 10=93.07%, 20=5.20%, 50=0.32%
cpu          : usr=0.78%, sys=4.42%, ctx=471975, majf=0, minf=0
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=165608,71031,0, short=0,0,0, dropped=0,0,0
latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
```

```
READ: bw=115MiB/s (121MB/s), 115MiB/s-115MiB/s (121MB/s-121MB/s), io=20.2GiB
(21.7GB), run=180009-180009msec
WRITE: bw=49.3MiB/s (51.7MB/s), 49.3MiB/s-49.3MiB/s (51.7MB/s-51.7MB/s),
io=8879MiB (9310MB), run=180009-180009msec

Disk stats (read/write):
sda: ios=165541/74271, merge=0/76, ticks=307094/72751, in_queue=357032,
util=78.18%
```

5.13 1.5G 固态FIO RAW写

```
seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1906: Tue Nov 24 16:58:14
2020
write: IOPS=125, BW=126MiB/s (132MB/s) (22.1GiB/180072msec)
clat (msec): min=23, max=553, avg=79.24, stdev=21.42
lat (msec): min=23, max=553, avg=79.52, stdev=21.42
clat percentiles (msec):
| 1.00th=[ 63], 5.00th=[ 71], 10.00th=[ 71], 20.00th=[ 78],
| 30.00th=[ 78], 40.00th=[ 79], 50.00th=[ 79], 60.00th=[ 79],
| 70.00th=[ 79], 80.00th=[ 79], 90.00th=[ 79], 95.00th=[ 86],
| 99.00th=[ 98], 99.50th=[ 258], 99.90th=[ 359], 99.95th=[ 367],
| 99.99th=[ 537]
bw ( KiB/s): min= 2072, max=16616, per=0.01%, avg=12944.93, stdev=1602.08
lat (msec) : 50=0.48%, 100=98.62%, 250=0.14%, 500=0.75%, 750=0.01%
cpu          : usr=0.38%, sys=0.39%, ctx=23915, majf=0, minf=3
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=0,22630,0, short=0,0,0, dropped=0,0,0
latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
WRITE: bw=126MiB/s (132MB/s), 126MiB/s-126MiB/s (132MB/s-132MB/s), io=22.1GiB
(23.7GB), run=180072-180072msec

Disk stats (read/write):
sda: ios=0/22629, merge=0/0, ticks=0/1785919, in_queue=1785792, util=100.00%
```

5.14 1.5G 固态FIO RAW读

```
seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=1926: Tue Nov 24 17:02:55
2020
read: IOPS=135, BW=136MiB/s (142MB/s) (23.8GiB/180069msec)
clat (msec): min=22, max=287, avg=73.70, stdev= 8.79
lat (msec): min=22, max=287, avg=73.70, stdev= 8.79
clat percentiles (msec):
| 1.00th=[ 60], 5.00th=[ 67], 10.00th=[ 74], 20.00th=[ 74],
| 30.00th=[ 74], 40.00th=[ 74], 50.00th=[ 74], 60.00th=[ 74],
| 70.00th=[ 74], 80.00th=[ 75], 90.00th=[ 75], 95.00th=[ 75],
| 99.00th=[ 82], 99.50th=[ 82], 99.90th=[ 260], 99.95th=[ 273],
| 99.99th=[ 289]
bw ( KiB/s): min= 6168, max=16616, per=0.01%, avg=13960.30, stdev=1047.08
```

```

    lat (msec) : 50=0.25%, 100=99.50%, 250=0.12%, 500=0.13%
cpu          : usr=0.05%, sys=0.49%, ctx=48939, majf=0, minf=2560
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwt: total=24419,0,0, short=0,0,0, dropped=0,0,0
    latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
    READ: bw=136MiB/s (142MB/s), 136MiB/s-136MiB/s (142MB/s-142MB/s), io=23.8GiB
(25.6GB), run=180069-180069msec

Disk stats (read/write):
    sda: ios=24385/0, merge=0/0, ticks=1787374/0, in_queue=1783850, util=100.00%
```

5.15 1.5G 固态FIO RAW读写

```

randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=1942: Tue Nov 24 17:07:04
2020
    read: IOPS=731, BW=91.5MiB/s (95.9MB/s) (16.1GiB/180010msec)
        clat (msec): min=2, max=52, avg= 8.33, stdev= 2.94
        lat (msec): min=2, max=52, avg= 8.33, stdev= 2.94
        clat percentiles (usec):
            | 1.00th=[ 4704], 5.00th=[ 5728], 10.00th=[ 6560], 20.00th=[ 7456],
            | 30.00th=[ 7520], 40.00th=[ 7584], 50.00th=[ 8384], 60.00th=[ 8512],
            | 70.00th=[ 8512], 80.00th=[ 8640], 90.00th=[ 9536], 95.00th=[10432],
            | 99.00th=[27520], 99.50th=[29568], 99.90th=[33024], 99.95th=[34048],
            | 99.99th=[42240]
        bw ( KiB/s): min= 3591, max=14592, per=0.01%, avg=9377.67, stdev=1486.42
    write: IOPS=314, BW=39.3MiB/s (41.2MB/s) (7069MiB/180010msec)
        clat (msec): min=1, max=55, avg=12.29, stdev= 4.42
        lat (msec): min=1, max=55, avg=12.33, stdev= 4.42
        clat percentiles (usec):
            | 1.00th=[ 5600], 5.00th=[ 8384], 10.00th=[ 8512], 20.00th=[ 9408],
            | 30.00th=[ 9664], 40.00th=[10432], 50.00th=[11328], 60.00th=[12224],
            | 70.00th=[13120], 80.00th=[14272], 90.00th=[17024], 95.00th=[19840],
            | 99.00th=[31360], 99.50th=[34048], 99.90th=[39680], 99.95th=[42752],
            | 99.99th=[48384]
        bw ( KiB/s): min= 1282, max= 7182, per=0.01%, avg=4024.96, stdev=830.57
        lat (msec) : 2=0.04%, 4=0.36%, 10=74.25%, 20=22.65%, 50=2.70%
        lat (msec) : 100=0.01%
    cpu          : usr=0.43%, sys=1.28%, ctx=322893, majf=0, minf=0
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
        submit   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
        complete : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
        issued rwt: total=131754,56549,0, short=0,0,0, dropped=0,0,0
        latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
    READ: bw=91.5MiB/s (95.9MB/s), 91.5MiB/s-91.5MiB/s (95.9MB/s-95.9MB/s),
io=16.1GiB (17.3GB), run=180010-180010msec
    WRITE: bw=39.3MiB/s (41.2MB/s), 39.3MiB/s-39.3MiB/s (41.2MB/s-41.2MB/s),
io=7069MiB (7412MB), run=180010-180010msec

Disk stats (read/write):
```

```
sda: ios=131642/56496, merge=0/0, ticks=1080048/687574, in_queue=1757585,
util=100.00%
```

5.16 1.5G 固态FIO EXT4写

```
seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1993: Tue Nov 24 17:26:43
2020
write: IOPS=104, BW=104MiB/s (110MB/s) (18.4GiB/180081msec)
  clat (msec): min=8, max=492, avg=95.33, stdev=19.82
    lat (msec): min=8, max=492, avg=95.66, stdev=19.82
  clat percentiles (msec):
    | 1.00th=[ 88], 5.00th=[ 90], 10.00th=[ 90], 20.00th=[ 90],
    | 30.00th=[ 90], 40.00th=[ 90], 50.00th=[ 90], 60.00th=[ 90],
    | 70.00th=[ 103], 80.00th=[ 104], 90.00th=[ 104], 95.00th=[ 104],
    | 99.00th=[ 119], 99.50th=[ 124], 99.90th=[ 445], 99.95th=[ 478],
    | 99.99th=[ 494]
  bw ( KiB/s): min= 4104, max=14539, per=0.01%, avg=10772.55, stdev=1206.94
    lat (msec) : 10=0.06%, 50=0.03%, 100=68.38%, 250=31.16%, 500=0.37%
  cpu          : usr=0.59%, sys=1.39%, ctx=38861, majf=0, minf=2
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwt: total=0,18815,0, short=0,0,0, dropped=0,0,0
    latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=104MiB/s (110MB/s), 104MiB/s-104MiB/s (110MB/s-110MB/s), io=18.4GiB
(19.7GB), run=180081-180081msec

Disk stats (read/write):
  sda: ios=1/21167, merge=0/221, ticks=12/204862, in_queue=199912, util=90.91%
```

5.17 1.5G 固态FIO EXT4读

```
seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=2024: Tue Nov 24 17:46:28
2020
read: IOPS=135, BW=136MiB/s (142MB/s) (23.8GiB/180067msec)
  clat (msec): min=21, max=294, avg=73.70, stdev=14.09
    lat (msec): min=21, max=294, avg=73.70, stdev=14.09
  clat percentiles (msec):
    | 1.00th=[ 60], 5.00th=[ 67], 10.00th=[ 67], 20.00th=[ 74],
    | 30.00th=[ 74], 40.00th=[ 74], 50.00th=[ 74], 60.00th=[ 74],
    | 70.00th=[ 74], 80.00th=[ 75], 90.00th=[ 75], 95.00th=[ 75],
    | 99.00th=[ 82], 99.50th=[ 237], 99.90th=[ 260], 99.95th=[ 273],
    | 99.99th=[ 273]
  bw ( KiB/s): min= 6168, max=18468, per=0.01%, avg=13981.93, stdev=1298.81
    lat (msec) : 50=0.38%, 100=98.94%, 250=0.34%, 500=0.34%
  cpu          : usr=0.05%, sys=1.15%, ctx=24777, majf=0, minf=2560
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwt: total=24418,0,0, short=0,0,0, dropped=0,0,0
```

```

    latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
    READ: bw=136MiB/s (142MB/s), 136MiB/s-136MiB/s (142MB/s-142MB/s), io=23.8GiB
(25.6GB), run=180067-180067msec

Disk stats (read/write):
    sda: ios=24401/12, merge=0/1, ticks=1784470/1085, in_queue=1781073,
util=100.00%

```

5.18 1.5G 固态FIO EXT4读写

```

randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=2249: Tue Nov 24 21:04:13
2020
    read: IOPS=549, BW=68.7MiB/s (72.1MB/s) (12.1GiB/180028msec)
        clat (msec): min=2, max=58, avg=12.57, stdev= 3.81
            lat (msec): min=2, max=58, avg=12.58, stdev= 3.81
        clat percentiles (usec):
            | 1.00th=[ 5536], 5.00th=[ 8512], 10.00th=[ 9920], 20.00th=[10816],
            | 30.00th=[11328], 40.00th=[11840], 50.00th=[12096], 60.00th=[12480],
            | 70.00th=[12864], 80.00th=[13376], 90.00th=[14272], 95.00th=[17280],
            | 99.00th=[28800], 99.50th=[29824], 99.90th=[41728], 99.95th=[43776],
            | 99.99th=[55552]
        bw ( KiB/s): min= 3855, max= 9767, per=0.01%, avg=7054.99, stdev=797.76
    write: IOPS=234, BW=29.4MiB/s (30.8MB/s) (5287MiB/180028msec)
        clat (msec): min=1, max=56, avg=12.99, stdev= 3.72
            lat (msec): min=1, max=56, avg=13.03, stdev= 3.72
        clat percentiles (usec):
            | 1.00th=[ 9792], 5.00th=[10432], 10.00th=[10816], 20.00th=[11328],
            | 30.00th=[11712], 40.00th=[11968], 50.00th=[12352], 60.00th=[12480],
            | 70.00th=[12864], 80.00th=[13376], 90.00th=[14016], 95.00th=[18304],
            | 99.00th=[28800], 99.50th=[29568], 99.90th=[42240], 99.95th=[43264],
            | 99.99th=[55040]
        bw ( KiB/s): min= 769, max= 5654, per=0.01%, avg=3014.60, stdev=724.07
        lat (msec) : 2=0.04%, 4=0.07%, 10=8.38%, 20=87.10%, 50=4.39%
        lat (msec) : 100=0.02%
    cpu           : usr=0.40%, sys=2.71%, ctx=282229, majf=0, minf=2
    IO depths     : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
        submit    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
        complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
        issued rwt: total=98969,42292,0, short=0,0,0, dropped=0,0,0
        latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
    READ: bw=68.7MiB/s (72.1MB/s), 68.7MiB/s-68.7MiB/s (72.1MB/s-72.1MB/s),
io=12.1GiB (12.0GB), run=180028-180028msec
    WRITE: bw=29.4MiB/s (30.8MB/s), 29.4MiB/s-29.4MiB/s (30.8MB/s-30.8MB/s),
io=5287MiB (5543MB), run=180028-180028msec

Disk stats (read/write):
    sda: ios=98936/44649, merge=0/74, ticks=336017/83716, in_queue=410934,
util=88.31%

```

5.19 6G 机械FIO RAW写

```
seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1685: Tue Nov 24 15:24:39
2020
write: IOPS=226, BW=227MiB/s (238MB/s) (39.8GiB/180037msec)
  clat (msec): min=7, max=6827, avg=43.82, stdev=120.95
    lat (msec): min=8, max=6827, avg=44.11, stdev=120.95
  clat percentiles (msec):
    | 1.00th=[ 21], 5.00th=[ 21], 10.00th=[ 21], 20.00th=[ 21],
    | 30.00th=[ 39], 40.00th=[ 39], 50.00th=[ 39], 60.00th=[ 40],
    | 70.00th=[ 40], 80.00th=[ 40], 90.00th=[ 56], 95.00th=[ 79],
    | 99.00th=[ 176], 99.50th=[ 243], 99.90th=[ 1057], 99.95th=[ 1582],
    | 99.99th=[ 6849]
  bw ( KiB/s): min= 2048, max=41541, per=0.01%, avg=24787.27, stdev=6090.58
    lat (msec) : 10=0.01%, 20=0.29%, 50=89.51%, 100=6.26%, 250=3.45%
    lat (msec) : 500=0.27%, 750=0.05%, 1000=0.03%, 2000=0.12%, >=2000=0.03%
  cpu          : usr=0.72%, sys=0.95%, ctx=42976, majf=0, minf=2
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwt: total=0,40780,0, short=0,0,0, dropped=0,0,0
    latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=227MiB/s (238MB/s), 227MiB/s-227MiB/s (238MB/s-238MB/s), io=39.8GiB
(42.8GB), run=180037-180037msec

Disk stats (read/write):
  sda: ios=0/52941, merge=0/0, ticks=0/2293956, in_queue=2293911, util=100.00%
```

5.20 6G 机械FIO RAW读

```
seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=1770: Tue Nov 24 15:42:48
2020
read: IOPS=430, BW=431MiB/s (451MB/s) (75.7GiB/180019msec)
  clat (msec): min=2, max=64, avg=23.19, stdev= 6.02
    lat (msec): min=2, max=64, avg=23.20, stdev= 6.02
  clat percentiles (usec):
    | 1.00th=[11456], 5.00th=[18816], 10.00th=[19072], 20.00th=[19072],
    | 30.00th=[19072], 40.00th=[19072], 50.00th=[19328], 60.00th=[23680],
    | 70.00th=[26240], 80.00th=[30080], 90.00th=[31104], 95.00th=[31872],
    | 99.00th=[41216], 99.50th=[44288], 99.90th=[51968], 99.95th=[55040],
    | 99.99th=[60672]
  bw ( KiB/s): min=32768, max=56195, per=0.01%, avg=44230.76, stdev=4049.64
    lat (msec) : 4=0.01%, 10=0.03%, 20=56.78%, 50=43.03%, 100=0.15%
  cpu          : usr=0.15%, sys=1.30%, ctx=162770, majf=0, minf=2560
  IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
    submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
    issued rwt: total=77513,0,0, short=0,0,0, dropped=0,0,0
    latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
```

```
READ: bw=431MiB/s (451MB/s), 431MiB/s-431MiB/s (451MB/s-451MB/s), io=75.7GiB
(81.3GB), run=180019-180019msec
```

Disk stats (read/write):

```
sda: ios=85192/0, merge=0/0, ticks=1943514/0, in_queue=1930157, util=99.68%
```

5.21 6G 机械FIO RAW读写

```
randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=1803: Tue Nov 24 15:58:47
2020
```

```
read: IOPS=154, BW=19.3MiB/s (20.2MB/s) (3467MiB/180082msec)
```

```
clat (msec): min=4, max=751, avg=63.50, stdev=66.70
```

```
lat (msec): min=4, max=751, avg=63.51, stdev=66.70
```

```
clat percentiles (msec):
```

```
| 1.00th=[ 8], 5.00th=[ 10], 10.00th=[ 13], 20.00th=[ 18],
| 30.00th=[ 24], 40.00th=[ 31], 50.00th=[ 41], 60.00th=[ 53],
| 70.00th=[ 71], 80.00th=[ 96], 90.00th=[ 147], 95.00th=[ 198],
| 99.00th=[ 322], 99.50th=[ 375], 99.90th=[ 529], 99.95th=[ 611],
| 99.99th=[ 709]
```

```
bw ( KiB/s): min= 256, max= 5130, per=0.01%, avg=1981.30, stdev=721.91
```

```
write: IOPS=65, BW=8366KiB/s (8567kB/s) (1471MiB/180082msec)
```

```
clat (usec): min=405, max=452016, avg=3178.54, stdev=4599.31
```

```
lat (usec): min=420, max=452043, avg=3212.31, stdev=4599.72
```

```
clat percentiles (usec):
```

```
| 1.00th=[ 588], 5.00th=[ 604], 10.00th=[ 660], 20.00th=[ 964],
| 30.00th=[ 1004], 40.00th=[ 2416], 50.00th=[ 3696], 60.00th=[ 4128],
| 70.00th=[ 4576], 80.00th=[ 4960], 90.00th=[ 5472], 95.00th=[ 5984],
| 99.00th=[ 7328], 99.50th=[ 8512], 99.90th=[11968], 99.95th=[12096],
| 99.99th=[12992]
```

```
bw ( KiB/s): min= 256, max= 4360, per=0.01%, avg=931.25, stdev=584.42
```

```
lat (usec) : 500=0.01%, 750=3.36%, 1000=5.39%
```

```
lat (msec) : 2=2.90%, 4=5.19%, 10=17.52%, 20=12.70%, 50=23.22%
```

```
lat (msec) : 100=16.37%, 250=11.60%, 500=1.63%, 750=0.10%, 1000=0.01%
```

```
cpu          : usr=0.09%, sys=0.29%, ctx=66008, majf=0, minf=0
```

```
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
```

```
submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
```

```
complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
```

```
issued rwt: total=27736,11770,0, short=0,0,0, dropped=0,0,0
```

```
latency     : target=0, window=0, percentile=100.00%, depth=1
```

Run status group 0 (all jobs):

```
READ: bw=19.3MiB/s (20.2MB/s), 19.3MiB/s-19.3MiB/s (20.2MB/s-20.2MB/s),
io=3467MiB (3635MB), run=180082-180082msec
```

```
WRITE: bw=8366KiB/s (8567kB/s), 8366KiB/s-8366KiB/s (8567kB/s-8567kB/s),
io=1471MiB (1543MB), run=180082-180082msec
```

Disk stats (read/write):

```
sda: ios=27707/11763, merge=0/0, ticks=1755482/36079, in_queue=1761213,
util=98.44%
```

5.22 6G 机械FIO EXT4写

```
seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1840: Tue Nov 24 16:08:16
2020
write: IOPS=179, BW=179MiB/s (188MB/s) (31.5GiB/180064msec)
  clat (msec): min=3, max=189, avg=55.47, stdev=17.47
    lat (msec): min=3, max=189, avg=55.77, stdev=17.47
  clat percentiles (msec):
    | 1.00th=[ 31], 5.00th=[ 32], 10.00th=[ 32], 20.00th=[ 45],
    | 30.00th=[ 47], 40.00th=[ 50], 50.00th=[ 55], 60.00th=[ 56],
    | 70.00th=[ 61], 80.00th=[ 67], 90.00th=[ 77], 95.00th=[ 85],
    | 99.00th=[ 119], 99.50th=[ 130], 99.90th=[ 155], 99.95th=[ 163],
    | 99.99th=[ 178]
  bw ( KiB/s): min=10240, max=26947, per=0.01%, avg=18461.36, stdev=2820.78
    lat (msec) : 4=0.04%, 20=0.02%, 50=40.76%, 100=56.69%, 250=2.49%
cpu           : usr=0.50%, sys=2.80%, ctx=65524, majf=0, minf=3
IO depths     : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
  submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  issued rwt: total=0,32261,0, short=0,0,0, dropped=0,0,0
  latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  WRITE: bw=179MiB/s (188MB/s), 179MiB/s-179MiB/s (188MB/s-188MB/s), io=31.5GiB
(33.8GB), run=180064-180064msec

Disk stats (read/write):
  sda: ios=1/38549, merge=0/243, ticks=11/196614, in_queue=189871, util=87.77%
```

5.23 6G 机械FIO EXT4读

```
seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=1858: Tue Nov 24 16:12:50
2020
read: IOPS=441, BW=442MiB/s (463MB/s) (77.7GiB/180024msec)
  clat (msec): min=2, max=120, avg=22.60, stdev= 7.44
    lat (msec): min=2, max=120, avg=22.60, stdev= 7.44
  clat percentiles (msec):
    | 1.00th=[ 15], 5.00th=[ 20], 10.00th=[ 20], 20.00th=[ 20],
    | 30.00th=[ 20], 40.00th=[ 20], 50.00th=[ 20], 60.00th=[ 20],
    | 70.00th=[ 21], 80.00th=[ 25], 90.00th=[ 32], 95.00th=[ 40],
    | 99.00th=[ 49], 99.50th=[ 55], 99.90th=[ 76], 99.95th=[ 82],
    | 99.99th=[ 103]
  bw ( KiB/s): min=28729, max=53461, per=0.01%, avg=45371.18, stdev=4391.10
    lat (msec) : 4=0.02%, 10=0.21%, 20=66.16%, 50=32.86%, 100=0.74%
    lat (msec) : 250=0.01%
cpu           : usr=0.17%, sys=3.64%, ctx=80690, majf=2, minf=2568
IO depths     : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
  submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  issued rwt: total=79560,0,0, short=0,0,0, dropped=0,0,0
  latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=442MiB/s (463MB/s), 442MiB/s-442MiB/s (463MB/s-463MB/s), io=77.7GiB
(83.4GB), run=180024-180024msec

Disk stats (read/write):
```

```
sda: ios=87461/2861, merge=0/42, ticks=1921401/67756, in_queue=1956591,
util=99.36%
```

5.24 6G 机械FIO EXT4读写

```
randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=1883: Tue Nov 24 16:20:06
2020
read: IOPS=89, BW=11.2MiB/s (11.7MB/s) (2008MiB/180056msec)
clat (msec): min=4, max=308, avg=79.74, stdev=38.98
lat (msec): min=4, max=308, avg=79.75, stdev=38.98
clat percentiles (msec):
| 1.00th=[ 14], 5.00th=[ 30], 10.00th=[ 41], 20.00th=[ 51],
| 30.00th=[ 59], 40.00th=[ 67], 50.00th=[ 73], 60.00th=[ 81],
| 70.00th=[ 89], 80.00th=[ 102], 90.00th=[ 130], 95.00th=[ 163],
| 99.00th=[ 210], 99.50th=[ 227], 99.90th=[ 258], 99.95th=[ 269],
| 99.99th=[ 297]
bw ( KiB/s): min= 256, max= 2313, per=0.01%, avg=1146.01, stdev=357.97
write: IOPS=37, BW=4850KiB/s (4967kB/s) (853MiB/180056msec)
clat (usec): min=656, max=307631, avg=75962.42, stdev=32347.37
lat (usec): min=691, max=307652, avg=75999.63, stdev=32347.41
clat percentiles (msec):
| 1.00th=[ 32], 5.00th=[ 43], 10.00th=[ 48], 20.00th=[ 55],
| 30.00th=[ 60], 40.00th=[ 64], 50.00th=[ 70], 60.00th=[ 75],
| 70.00th=[ 81], 80.00th=[ 89], 90.00th=[ 111], 95.00th=[ 151],
| 99.00th=[ 198], 99.50th=[ 217], 99.90th=[ 245], 99.95th=[ 249],
| 99.99th=[ 310]
bw ( KiB/s): min= 256, max= 1542, per=0.01%, avg=552.43, stdev=264.13
lat (usec) : 750=0.04%, 1000=0.01%
lat (msec) : 10=0.38%, 20=1.22%, 50=15.49%, 100=64.23%, 250=18.50%
lat (msec) : 500=0.12%
cpu          : usr=0.07%, sys=0.45%, ctx=46120, majf=0, minf=0
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=16063,6823,0, short=0,0,0, dropped=0,0,0
latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=11.2MiB/s (11.7MB/s), 11.2MiB/s-11.2MiB/s (11.7MB/s-11.7MB/s),
io=2008MiB (2105MB), run=180056-180056msec
  WRITE: bw=4850KiB/s (4967kB/s), 4850KiB/s-4850KiB/s (4967kB/s-4967kB/s),
io=853MiB (894MB), run=180056-180056msec

Disk stats (read/write):
  sda: ios=16046/9823, merge=0/76, ticks=458841/35050, in_queue=464929,
util=93.20%
```

5.25 3G 机械FIO RAW写

```
seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1913: Tue Nov 24 16:33:27
2020
write: IOPS=184, BW=184MiB/s (193MB/s) (32.4GiB/180036msec)
```

```

clat (msec): min=4, max=470, avg=53.89, stdev=11.89
lat (msec): min=5, max=470, avg=54.17, stdev=11.89
clat percentiles (msec):
| 1.00th=[ 39], 5.00th=[ 44], 10.00th=[ 45], 20.00th=[ 45],
| 30.00th=[ 46], 40.00th=[ 47], 50.00th=[ 55], 60.00th=[ 57],
| 70.00th=[ 60], 80.00th=[ 64], 90.00th=[ 67], 95.00th=[ 74],
| 99.00th=[ 75], 99.50th=[ 75], 99.90th=[ 86], 99.95th=[ 104],
| 99.99th=[ 469]
bw ( KiB/s): min= 4096, max=20770, per=0.01%, avg=18986.94, stdev=1492.77
lat (msec) : 10=0.01%, 20=0.01%, 50=46.60%, 100=53.33%, 250=0.02%
lat (msec) : 500=0.03%
cpu          : usr=0.57%, sys=0.57%, ctx=37035, majf=0, minf=3
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=0,33204,0, short=0,0,0, dropped=0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
WRITE: bw=184MiB/s (193MB/s), 184MiB/s-184MiB/s (193MB/s-193MB/s), io=32.4GiB
(34.8GB), run=180036-180036msec

Disk stats (read/write):
sda: ios=0/33205, merge=0/895, ticks=0/1777898, in_queue=1746829, util=98.40%

```

5.26 3G 机械FIO RAW读

```

seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=1843: Sat Nov 14 02:12:49
2020
read: IOPS=262, BW=263MiB/s (276MB/s) (46.2GiB/180036msec)
clat (msec): min=9, max=79, avg=38.01, stdev= 3.31
lat (msec): min=9, max=79, avg=38.02, stdev= 3.31
clat percentiles (usec):
| 1.00th=[36608], 5.00th=[36608], 10.00th=[36608], 20.00th=[36608],
| 30.00th=[36608], 40.00th=[37120], 50.00th=[37120], 60.00th=[37120],
| 70.00th=[37120], 80.00th=[37120], 90.00th=[43264], 95.00th=[46848],
| 99.00th=[47872], 99.50th=[48384], 99.90th=[62720], 99.95th=[70144],
| 99.99th=[77312]
bw ( KiB/s): min=20480, max=29079, per=0.01%, avg=27047.76, stdev=1790.16
lat (msec) : 10=0.01%, 20=0.06%, 50=99.71%, 100=0.22%
cpu          : usr=0.09%, sys=0.76%, ctx=99482, majf=0, minf=2560
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=47316,0,0, short=0,0,0, dropped=0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
READ: bw=263MiB/s (276MB/s), 263MiB/s-263MiB/s (276MB/s-276MB/s), io=46.2GiB
(49.6GB), run=180036-180036msec

Disk stats (read/write):
sda: ios=51994/0, merge=0/0, ticks=1951402/0, in_queue=1948913, util=100.00%

```

5.27 3G 机械FIO RAW读写

```
randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=1864: Sat Nov 14 02:19:47
2020
read: IOPS=154, BW=19.3MiB/s (20.3MB/s) (3486MiB/180168msec)
  clat (msec): min=4, max=862, avg=63.07, stdev=64.99
    lat (msec): min=4, max=862, avg=63.08, stdev=64.99
  clat percentiles (msec):
    | 1.00th=[ 8], 5.00th=[ 10], 10.00th=[ 13], 20.00th=[ 18],
    | 30.00th=[ 24], 40.00th=[ 32], 50.00th=[ 41], 60.00th=[ 53],
    | 70.00th=[ 70], 80.00th=[ 95], 90.00th=[ 143], 95.00th=[ 194],
    | 99.00th=[ 318], 99.50th=[ 371], 99.90th=[ 498], 99.95th=[ 545],
    | 99.99th=[ 619]
  bw ( KiB/s): min= 256, max= 4626, per=0.01%, avg=1991.20, stdev=714.46
write: IOPS=65, BW=8400KiB/s (8601kB/s) (1478MiB/180168msec)
  clat (usec): min=626, max=13245, avg=3373.21, stdev=2033.05
    lat (usec): min=638, max=13266, avg=3408.54, stdev=2034.17
  clat percentiles (usec):
    | 1.00th=[ 836], 5.00th=[ 852], 10.00th=[ 884], 20.00th=[ 1176],
    | 30.00th=[ 1224], 40.00th=[ 2736], 50.00th=[ 3952], 60.00th=[ 4384],
    | 70.00th=[ 4768], 80.00th=[ 5152], 90.00th=[ 5728], 95.00th=[ 6240],
    | 99.00th=[ 7968], 99.50th=[ 8896], 99.90th=[12096], 99.95th=[12480],
    | 99.99th=[13248]
  bw ( KiB/s): min= 256, max= 4369, per=0.01%, avg=929.73, stdev=579.06
    lat (usec) : 750=0.03%, 1000=4.45%
    lat (msec) : 2=7.00%, 4=3.64%, 10=18.32%, 20=12.96%, 50=24.15%
    lat (msec) : 100=16.50%, 250=11.29%, 500=1.60%, 750=0.06%, 1000=0.01%
cpu           : usr=0.10%, sys=0.27%, ctx=68262, majf=0, minf=0
IO depths     : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
  submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  issued rwts: total=27889,11823,0, short=0,0,0, dropped=0,0,0
  latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=19.3MiB/s (20.3MB/s), 19.3MiB/s-19.3MiB/s (20.3MB/s-20.3MB/s),
io=3486MiB (3655MB), run=180168-180168msec
  WRITE: bw=8400KiB/s (8601kB/s), 8400KiB/s-8400KiB/s (8601kB/s-8601kB/s),
io=1478MiB (1550MB), run=180168-180168msec

Disk stats (read/write):
  sda: ios=27886/11823, merge=0/0, ticks=1754898/38561, in_queue=1792906,
util=100.00%
```

5.28 3G 机械FIO EXT4写

```
seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1945: Tue Nov 24 16:40:06
2020
write: IOPS=136, BW=136MiB/s (143MB/s) (23.0GiB/180105msec)
  clat (msec): min=4, max=242, avg=73.01, stdev=16.29
    lat (msec): min=4, max=242, avg=73.31, stdev=16.29
  clat percentiles (msec):
    | 1.00th=[ 50], 5.00th=[ 58], 10.00th=[ 63], 20.00th=[ 63],
```

```

| 30.00th=[ 64], 40.00th=[ 64], 50.00th=[ 70], 60.00th=[ 72],
| 70.00th=[ 75], 80.00th=[ 82], 90.00th=[ 94], 95.00th=[ 101],
| 99.00th=[ 135], 99.50th=[ 157], 99.90th=[ 188], 99.95th=[ 194],
| 99.99th=[ 235]
bw ( KiB/s): min= 8192, max=18580, per=0.01%, avg=14107.55, stdev=1484.39
lat (msec) : 10=0.04%, 20=0.02%, 50=1.72%, 100=93.07%, 250=5.15%
cpu          : usr=0.50%, sys=2.03%, ctx=50021, majf=0, minf=0
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=0,24546,0, short=0,0,0, dropped=0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
WRITE: bw=136MiB/s (143MB/s), 136MiB/s-136MiB/s (143MB/s-143MB/s), io=23.0GiB
(25.7GB), run=180105-180105msec

Disk stats (read/write):
sda: ios=1/29607, merge=0/236, ticks=9/216452, in_queue=205944, util=88.47%
```

5.29 3G 机械FIO EXT4读

```

seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=1984: Tue Nov 24 16:55:43
2020
read: IOPS=250, BW=250MiB/s (262MB/s) (43.0GiB/180035msec)
clat (msec): min=7, max=119, avg=39.94, stdev= 8.79
lat (msec): min=7, max=119, avg=39.94, stdev= 8.79
clat percentiles (msec):
| 1.00th=[ 19], 5.00th=[ 34], 10.00th=[ 37], 20.00th=[ 37],
| 30.00th=[ 37], 40.00th=[ 37], 50.00th=[ 37], 60.00th=[ 38],
| 70.00th=[ 38], 80.00th=[ 45], 90.00th=[ 49], 95.00th=[ 57],
| 99.00th=[ 76], 99.50th=[ 81], 99.90th=[ 98], 99.95th=[ 105],
| 99.99th=[ 120]
bw ( KiB/s): min=18432, max=30781, per=0.01%, avg=25692.11, stdev=1972.95
lat (msec) : 10=0.02%, 20=1.36%, 50=89.64%, 100=8.90%, 250=0.07%
cpu          : usr=0.10%, sys=2.10%, ctx=45599, majf=0, minf=2560
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=45037,0,0, short=0,0,0, dropped=0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
READ: bw=250MiB/s (262MB/s), 250MiB/s-250MiB/s (262MB/s-262MB/s), io=43.0GiB
(47.2GB), run=180035-180035msec

Disk stats (read/write):
sda: ios=44989/2325, merge=0/40, ticks=1771943/96460, in_queue=1834482,
util=99.18%
```

5.30 3G 机械FIO EXT4读写

```

randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=1821: Sat Nov 14 02:04:53
2020
read: IOPS=97, BW=12.1MiB/s (12.7MB/s) (2187MiB/180115msec)
clat (msec): min=5, max=227, avg=72.72, stdev=27.60
lat (msec): min=5, max=227, avg=72.72, stdev=27.60
clat percentiles (msec):
| 1.00th=[ 15], 5.00th=[ 31], 10.00th=[ 41], 20.00th=[ 51],
| 30.00th=[ 59], 40.00th=[ 65], 50.00th=[ 72], 60.00th=[ 79],
| 70.00th=[ 85], 80.00th=[ 93], 90.00th=[ 106], 95.00th=[ 120],
| 99.00th=[ 155], 99.50th=[ 174], 99.90th=[ 198], 99.95th=[ 206],
| 99.99th=[ 223]
bw ( KiB/s): min= 256, max= 2313, per=0.01%, avg=1247.62, stdev=352.57
write: IOPS=41, BW=5319KiB/s (5447kB/s) (936MiB/180115msec)
clat (usec): min=817, max=197771, avg=70451.12, stdev=20182.61
lat (usec): min=842, max=197809, avg=70490.22, stdev=20182.64
clat percentiles (msec):
| 1.00th=[ 35], 5.00th=[ 43], 10.00th=[ 48], 20.00th=[ 55],
| 30.00th=[ 60], 40.00th=[ 65], 50.00th=[ 70], 60.00th=[ 74],
| 70.00th=[ 79], 80.00th=[ 85], 90.00th=[ 94], 95.00th=[ 102],
| 99.00th=[ 143], 99.50th=[ 159], 99.90th=[ 180], 99.95th=[ 184],
| 99.99th=[ 198]
bw ( KiB/s): min= 256, max= 1799, per=0.01%, avg=583.17, stdev=285.13
lat (usec) : 1000=0.03%
lat (msec) : 2=0.02%, 10=0.36%, 20=1.10%, 50=16.03%, 100=70.92%
lat (msec) : 250=11.55%
cpu          : usr=0.09%, sys=0.47%, ctx=50401, majf=0, minf=0
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit      : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete    : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=17494,7485,0, short=0,0,0, dropped=0,0,0
latency     : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=12.1MiB/s (12.7MB/s), 12.1MiB/s-12.1MiB/s (12.7MB/s-12.7MB/s),
io=2187MiB (2293MB), run=180115-180115msec
  WRITE: bw=5319KiB/s (5447kB/s), 5319KiB/s-5319KiB/s (5447kB/s-5447kB/s),
io=936MiB (981MB), run=180115-180115msec

Disk stats (read/write):
  sda: ios=17490/7557, merge=0/36, ticks=433099/10178, in_queue=436335,
util=97.52%

```

5.31 1.5G 机械FIO RAW写

```

seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=1954: Sat Nov 14 02:52:00
2020
write: IOPS=112, BW=112MiB/s (118MB/s) (19.7GiB/180085msec)
clat (msec): min=8, max=412, avg=88.71, stdev=11.99
lat (msec): min=8, max=412, avg=89.01, stdev=11.99
clat percentiles (msec):
| 1.00th=[ 77], 5.00th=[ 78], 10.00th=[ 78], 20.00th=[ 78],
| 30.00th=[ 87], 40.00th=[ 88], 50.00th=[ 90], 60.00th=[ 91],
| 70.00th=[ 92], 80.00th=[ 95], 90.00th=[ 100], 95.00th=[ 106],
| 99.00th=[ 115], 99.50th=[ 133], 99.90th=[ 206], 99.95th=[ 215],
| 99.99th=[ 396]

```

```

bw ( KiB/s): min= 6156, max=14422, per=0.01%, avg=11580.72, stdev=1067.37
lat (msec) : 10=0.01%, 20=0.01%, 50=0.05%, 100=92.78%, 250=7.11%
lat (msec) : 500=0.05%
cpu          : usr=0.36%, sys=0.36%, ctx=24231, majf=0, minf=2
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=0,20214,0, short=0,0,0, dropped=0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
WRITE: bw=112MiB/s (118MB/s), 112MiB/s-112MiB/s (118MB/s-118MB/s), io=19.7GiB
(21.2GB), run=180085-180085msec

Disk stats (read/write):
sda: ios=0/20284, merge=0/747, ticks=0/1788377, in_queue=1783124, util=99.91%

```

5.32 1.5G 机械FIO RAW读

```

seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=1973: Sat Nov 14 02:57:32
2020
read: IOPS=135, BW=136MiB/s (143MB/s) (23.9GiB/180067msec)
clat (usec): min=22151, max=132908, avg=73528.77, stdev=763.20
lat (usec): min=22154, max=132909, avg=73531.63, stdev=763.17
clat percentiles (usec):
| 1.00th=[73216], 5.00th=[73216], 10.00th=[73216], 20.00th=[73216],
| 30.00th=[73216], 40.00th=[73216], 50.00th=[73216], 60.00th=[73216],
| 70.00th=[73216], 80.00th=[73216], 90.00th=[73216], 95.00th=[73216],
| 99.00th=[73216], 99.50th=[73216], 99.90th=[74240], 99.95th=[74240],
| 99.99th=[96768]
bw ( KiB/s): min=12312, max=14451, per=0.01%, avg=13991.68, stdev=831.22
lat (msec) : 50=0.02%, 100=99.98%, 250=0.01%
cpu          : usr=0.05%, sys=0.43%, ctx=49106, majf=0, minf=2560
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=24473,0,0, short=0,0,0, dropped=0,0,0
latency      : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
READ: bw=136MiB/s (143MB/s), 136MiB/s-136MiB/s (143MB/s-143MB/s), io=23.9GiB
(25.7GB), run=180067-180067msec

Disk stats (read/write):
sda: ios=24444/10, merge=0/0, ticks=1788949/42, in_queue=1788252, util=100.00%

```

5.33 1.5G 机械FIO RAW读写

```

randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=1991: Sat Nov 14 03:02:31
2020
read: IOPS=153, BW=19.2MiB/s (20.1MB/s) (3454MiB/180115msec)
clat (msec): min=5, max=939, avg=63.43, stdev=63.35

```

```

    lat (msec): min=5, max=939, avg=63.44, stdev=63.35
    clat percentiles (msec):
      | 1.00th=[ 10], 5.00th=[ 13], 10.00th=[ 15], 20.00th=[ 20],
      | 30.00th=[ 26], 40.00th=[ 33], 50.00th=[ 42], 60.00th=[ 54],
      | 70.00th=[ 70], 80.00th=[ 95], 90.00th=[ 141], 95.00th=[ 188],
      | 99.00th=[ 306], 99.50th=[ 363], 99.90th=[ 510], 99.95th=[ 562],
      | 99.99th=[ 758]
    bw ( KiB/s): min= 256, max= 4360, per=0.01%, avg=1970.27, stdev=665.76
    write: IOPS=65, BW=8323KiB/s (8523kB/s) (1464MiB/180115msec)
    clat (msec): min=1, max=13, avg= 3.89, stdev= 2.04
    lat (msec): min=1, max=13, avg= 3.92, stdev= 2.04
    clat percentiles (usec):
      | 1.00th=[ 1352], 5.00th=[ 1368], 10.00th=[ 1400], 20.00th=[ 1448],
      | 30.00th=[ 1880], 40.00th=[ 3440], 50.00th=[ 4448], 60.00th=[ 4896],
      | 70.00th=[ 5280], 80.00th=[ 5664], 90.00th=[ 6176], 95.00th=[ 6816],
      | 99.00th=[ 8640], 99.50th=[ 9408], 99.90th=[11840], 99.95th=[12736],
      | 99.99th=[12992]
    bw ( KiB/s): min= 256, max= 3598, per=0.01%, avg=924.03, stdev=563.43
    lat (msec) : 2=10.35%, 4=2.91%, 10=17.53%, 20=13.25%, 50=25.97%
    lat (msec) : 100=17.03%, 250=11.43%, 500=1.45%, 750=0.07%, 1000=0.01%
    cpu          : usr=0.09%, sys=0.28%, ctx=67701, majf=0, minf=0
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      complete   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
      issued rwt: total=27634,11712,0, short=0,0,0, dropped=0,0,0
      latency    : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
    READ: bw=19.2MiB/s (20.1MB/s), 19.2MiB/s-19.2MiB/s (20.1MB/s-20.1MB/s),
    io=3454MiB (3622MB), run=180115-180115msec
    WRITE: bw=8323KiB/s (8523kB/s), 8323KiB/s-8323KiB/s (8523kB/s-8523kB/s),
    io=1464MiB (1535MB), run=180115-180115msec

Disk stats (read/write):
    sda: ios=27633/11722, merge=0/0, ticks=1749080/43881, in_queue=1791393,
    util=100.00%

```

5.34 1.5G 机械FIO EXT4写

```

seq_100write_1024k: (groupid=0, jobs=10): err= 0: pid=2019: Sat Nov 14 03:10:51
2020
    write: IOPS=86, BW=86.0MiB/s (91.2MB/s) (15.3GiB/180155msec)
    clat (msec): min=8, max=245, avg=114.59, stdev=16.52
    lat (msec): min=9, max=245, avg=114.92, stdev=16.52
    clat percentiles (msec):
      | 1.00th=[ 89], 5.00th=[ 95], 10.00th=[ 103], 20.00th=[ 105],
      | 30.00th=[ 106], 40.00th=[ 111], 50.00th=[ 114], 60.00th=[ 115],
      | 70.00th=[ 116], 80.00th=[ 122], 90.00th=[ 131], 95.00th=[ 139],
      | 99.00th=[ 190], 99.50th=[ 198], 99.90th=[ 225], 99.95th=[ 231],
      | 99.99th=[ 245]
    bw ( KiB/s): min= 6144, max=12487, per=0.01%, avg=8981.66, stdev=1120.68
    lat (msec) : 10=0.03%, 20=0.02%, 50=0.01%, 100=7.19%, 250=92.75%
    cpu          : usr=0.34%, sys=1.31%, ctx=32281, majf=0, minf=0
    IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
      submit     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%

```

```
complete : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
issued rwt: total=0,15666,0, short=0,0,0, dropped=0,0,0
latency   : target=0, window=0, percentile=100.00%, depth=1
```

Run status group 0 (all jobs):

```
WRITE: bw=86.0MiB/s (91.2MB/s), 86.0MiB/s-86.0MiB/s (91.2MB/s-91.2MB/s),
io=15.3GiB (16.4GB), run=180155-180155msec
```

Disk stats (read/write):

```
sda: ios=1/19046, merge=0/214, ticks=0/217398, in_queue=211239, util=93.40%
```

5.35 1.5G 机械FIO EXT4读

```
seq_100read_1024k: (groupid=0, jobs=10): err= 0: pid=2042: Sat Nov 14 03:18:10
2020
```

```
read: IOPS=131, BW=131MiB/s (138MB/s) (23.1GiB/180070msec)
```

```
clat (msec): min=15, max=157, avg=76.17, stdev= 7.91
```

```
lat (msec): min=15, max=157, avg=76.17, stdev= 7.91
```

```
clat percentiles (msec):
```

```
| 1.00th=[ 60], 5.00th=[ 74], 10.00th=[ 74], 20.00th=[ 74],
| 30.00th=[ 74], 40.00th=[ 74], 50.00th=[ 74], 60.00th=[ 74],
| 70.00th=[ 74], 80.00th=[ 74], 90.00th=[ 89], 95.00th=[ 89],
| 99.00th=[ 100], 99.50th=[ 117], 99.90th=[ 141], 99.95th=[ 149],
| 99.99th=[ 157]
```

```
bw ( KiB/s): min=10240, max=16616, per=0.01%, avg=13491.35, stdev=1056.88
```

```
lat (msec) : 20=0.01%, 50=0.50%, 100=98.46%, 250=1.03%
```

```
cpu          : usr=0.05%, sys=1.04%, ctx=24344, majf=0, minf=2564
```

```
IO depths    : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
```

```
submit       : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
```

```
complete     : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
```

```
issued rwt: total=23626,0,0, short=0,0,0, dropped=0,0,0
```

```
latency      : target=0, window=0, percentile=100.00%, depth=1
```

Run status group 0 (all jobs):

```
READ: bw=131MiB/s (138MB/s), 131MiB/s-131MiB/s (138MB/s-138MB/s), io=23.1GiB
(24.8GB), run=180070-180070msec
```

Disk stats (read/write):

```
sda: ios=23614/1575, merge=0/35, ticks=1785211/125916, in_queue=1884301,
util=100.00%
```

5.36 1.5G 机械FIO EXT4读写

```
randrw_70read_128k: (groupid=0, jobs=10): err= 0: pid=2062: Sat Nov 14 03:26:19
2020
```

```
read: IOPS=84, BW=10.6MiB/s (11.1MB/s) (1913MiB/180072msec)
```

```
clat (msec): min=6, max=297, avg=83.62, stdev=32.09
```

```
lat (msec): min=6, max=298, avg=83.62, stdev=32.09
```

```
clat percentiles (msec):
```

```
| 1.00th=[ 17], 5.00th=[ 35], 10.00th=[ 46], 20.00th=[ 59],
| 30.00th=[ 68], 40.00th=[ 75], 50.00th=[ 81], 60.00th=[ 89],
| 70.00th=[ 96], 80.00th=[ 108], 90.00th=[ 126], 95.00th=[ 141],
```

```

    | 99.00th=[ 176], 99.50th=[ 192], 99.90th=[ 235], 99.95th=[ 245],
    | 99.99th=[ 285]
bw ( KiB/s): min= 256, max= 2565, per=0.01%, avg=1092.47, stdev=338.82
write: IOPS=35, BW=4604KiB/s (4715kB/s) (810MiB/180072msec)
  clat (msec): min=1, max=237, avg=80.21, stdev=23.97
  lat (msec): min=1, max=237, avg=80.25, stdev=23.97
  clat percentiles (msec):
    | 1.00th=[ 41], 5.00th=[ 50], 10.00th=[ 56], 20.00th=[ 62],
    | 30.00th=[ 68], 40.00th=[ 72], 50.00th=[ 76], 60.00th=[ 81],
    | 70.00th=[ 88], 80.00th=[ 97], 90.00th=[ 114], 95.00th=[ 126],
    | 99.00th=[ 153], 99.50th=[ 172], 99.90th=[ 202], 99.95th=[ 227],
    | 99.99th=[ 239]
bw ( KiB/s): min= 256, max= 1542, per=0.01%, avg=527.19, stdev=253.77
  lat (msec) : 2=0.05%, 10=0.13%, 20=0.94%, 50=9.42%, 100=65.63%
  lat (msec) : 250=23.81%, 500=0.02%
cpu      : usr=0.07%, sys=0.44%, ctx=43966, majf=0, minf=1
IO depths : 1=100.0%, 2=0.0%, 4=0.0%, 8=0.0%, 16=0.0%, 32=0.0%, >=64=0.0%
  submit   : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  complete  : 0=0.0%, 4=100.0%, 8=0.0%, 16=0.0%, 32=0.0%, 64=0.0%, >=64=0.0%
  issued rwt: total=15305,6477,0, short=0,0,0, dropped=0,0,0
  latency   : target=0, window=0, percentile=100.00%, depth=1

Run status group 0 (all jobs):
  READ: bw=10.6MiB/s (11.1MB/s), 10.6MiB/s-10.6MiB/s (11.1MB/s-11.1MB/s),
io=1913MiB (2006MB), run=180072-180072msec
  WRITE: bw=4604KiB/s (4715kB/s), 4604KiB/s-4604KiB/s (4715kB/s-4715kB/s),
io=810MiB (849MB), run=180072-180072msec

Disk stats (read/write):
  sda: ios=15301/8281, merge=0/74, ticks=465486/44059, in_queue=500849,
util=97.88%

```